

Chapter 5 Regression-Based Classification

Austin R.J. Downey

Regression-Based Classification

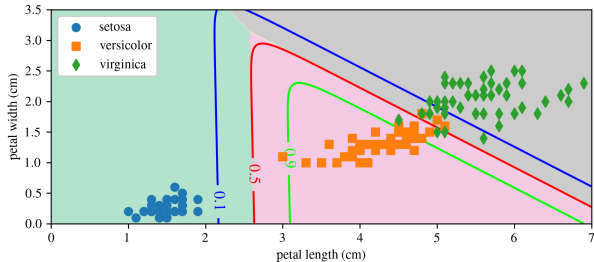
Regression models can often be adapted for classification.

Examples:

- ▶ Linear Regression $\rightarrow$ Logistic Regression
- ▶ Logistic Regression $\rightarrow$ Softmax Regression

Advantages:

- ▶ Interpretable models
- ▶ Familiar hyperparameters
- ▶ Probabilistic predictions



Classification regions for the three Iris species.

Logistic Regression

Logistic Regression estimates the probability that an input belongs to a class.

Predicted probability:

$$\hat{p} = h_{\theta}(X) = \sigma(\boldsymbol{\theta}^{\top} X)$$

Classification rule:

$$\hat{y} = \begin{cases} 0 & \hat{p} < 0.5 \\ 1 & \hat{p} \geq 0.5 \end{cases}$$

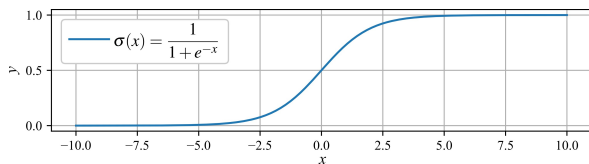
- ▶ Binary classifier
- ▶ Output is a probability
- ▶ Threshold typically set to 50%

The Sigmoid Function

The sigmoid function maps any real number to the interval $(0, 1)$:

:contentReference[oaicite:0]index=0

- ▶ S-shaped curve
- ▶ Converts scores into probabilities
- ▶ Central component of Logistic Regression



Sigmoid function.

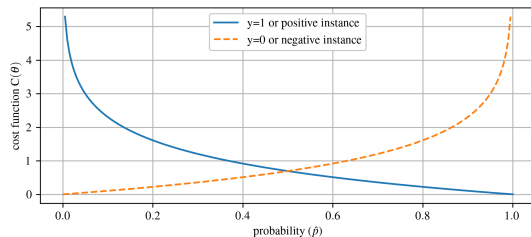
Classification Cost Function

Training Logistic Regression requires a cost function that:

- ▶ Rewards correct predictions
- ▶ Penalizes incorrect predictions
- ▶ Produces probabilities near:
 - ▶ 1 for positive examples
 - ▶ 0 for negative examples

Observations:

- ▶ Large penalty when a confident prediction is wrong
- ▶ Small penalty when a confident prediction is correct



Classification cost function.

Training Logistic Regression

Cost for a single sample:

$$C(\theta) = \begin{cases} -\log(\hat{p}) & y = 1 \\ -\log(1 - \hat{p}) & y = 0 \end{cases}$$

Overall cost (log-loss):

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{p}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]$$

- ▶ No closed-form solution
- ▶ Train using Gradient Descent
- ▶ Convex cost function
- ▶ Guaranteed global minimum

Data: Iris Flower Dataset

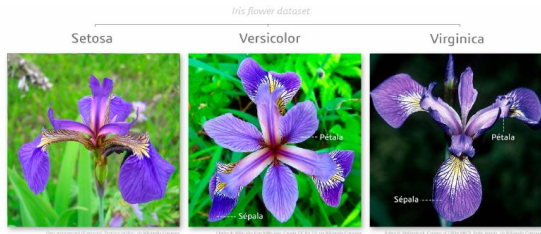
The Iris dataset contains measurements from:

- ▶ 150 flowers
- ▶ 3 species:
 - ▶ Iris-Setosa
 - ▶ Iris-Versicolor
 - ▶ Iris-Virginica

Features:

- ▶ Sepal length
- ▶ Sepal width
- ▶ Petal length
- ▶ Petal width

A classic classification dataset.



Three Iris flower species.

Data: Iris Measurements

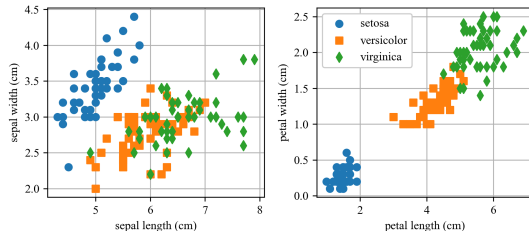
The four measured features can be used to classify flower species.

Observations:

- ▶ Setosa is easily separated
- ▶ Versicolor and Virginica overlap
- ▶ Petal measurements provide better separation than sepal measurements

Goal:

- ▶ Predict species from measurements



Sepal and petal measurements for the Iris dataset.

Example 5.1: Iris Dataset Exploration

- ▶ Load the Iris dataset from Scikit-Learn
- ▶ Examine feature and label information
- ▶ Visualize sepal and petal measurements
- ▶ Explore relationships between the three species

Demonstrates:

- ▶ Classification datasets
- ▶ Feature exploration
- ▶ Data visualization
- ▶ Class separation

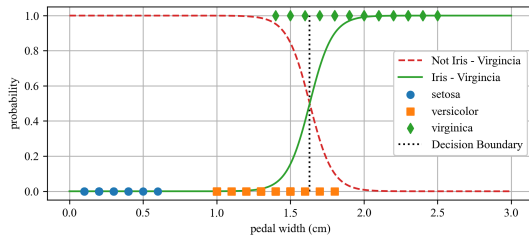
1-D Decision Boundaries

Logistic Regression can classify
Iris-Virginica using a single feature:

Petal Width

Observations:

- ▶ Below 1 cm:
 - ▶ Confidently not Virginica
- ▶ Above 2 cm:
 - ▶ Confidently Virginica
- ▶ Between:
 - ▶ Model uncertainty



1-D decision boundary.

Decision Boundary

A Logistic Regression model predicts probabilities:

$$\hat{p} = P(y = 1|X)$$

where P denotes probability. In this case, $P(y = 1|X)$ means the probability that the sample belongs to class 1, given the input features X .

The classification boundary occurs when:

$$\hat{p} = 0.5$$

For the Iris dataset:

- Decision boundary occurs near:

$$\text{Petal Width} \approx 1.6 \text{ cm}$$

- Above the boundary: predict Iris-Virginica; below the boundary: predict not Iris-Virginica.

Example 5.2: 1D Logistic Regression

- ▶ Train a Logistic Regression model using:
 - ▶ Petal width only
- ▶ Estimate:
 - ▶ Probability of Iris-Virginica
- ▶ Visualize:
 - ▶ Predicted probabilities
 - ▶ 50% decision boundary
- ▶ Demonstrates how probabilities become class predictions

2-D Decision Boundaries

Using two features:

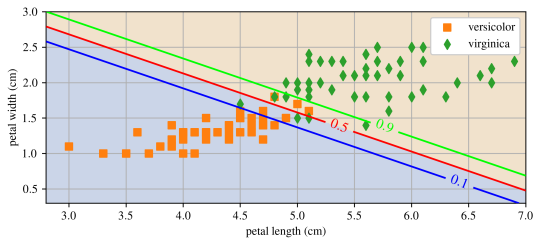
- ▶ Petal length
- ▶ Petal width

The model predicts:

$$P(\text{Virginica}|X)$$

Features:

- ▶ Dashed line:
 - ▶ 50% decision boundary
- ▶ Contours:
 - ▶ Equal probability levels
- ▶ Upper-right region:
 - ▶ High confidence Virginica



2-D decision boundary and probability contours.

Example 5.3: 2D Decision Boundary

- ▶ Train a Logistic Regression model using:
 - ▶ Petal length
 - ▶ Petal width
- ▶ Visualize:
 - ▶ Classification regions
 - ▶ Decision boundary
 - ▶ Probability contours
- ▶ Demonstrates how Logistic Regression separates classes in a two-dimensional feature space

Softmax Regression

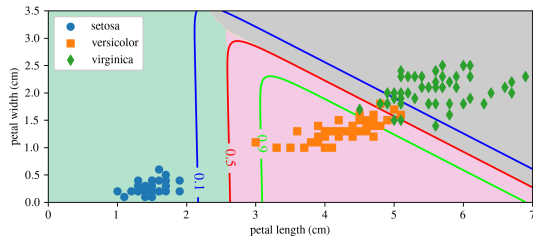
Softmax Regression extends Logistic Regression to multiple classes.

- ▶ No need to train several binary classifiers
- ▶ Works when each input belongs to exactly one class
- ▶ Also called Multinomial Logistic Regression

For an input vector $\mathbf{x}$, the model computes a score for each class.

Example:

- ▶ Classify all three Iris species simultaneously



Softmax classification of the three Iris species.

Class Scores and Probabilities

For class k , the score is

$$s_k(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\theta}^{(k)}$$

The softmax function converts scores into probabilities:

$$\hat{p}_k = \frac{\exp(s_k(\mathbf{x}))}{\sum_{j=1}^K \exp(s_j(\mathbf{x}))}$$

- ▶ $\mathbf{s}(\mathbf{x})$ is the vector of class scores
- ▶ K is the number of classes
- ▶ All predicted probabilities sum to 1

Prediction Rule

Softmax Regression predicts the class with the largest probability:

$$\hat{y} = \operatorname{argmax}_k \hat{p}_k = \operatorname{argmax}_k s_k(\mathbf{x}) = \operatorname{argmax}_k ((\boldsymbol{\theta}^{(k)})^\top \mathbf{x})$$

Note

The argmax operator returns the index of the largest value.

- ▶ Softmax Regression is multiclass, not multioutput
- ▶ It predicts one class at a time
- ▶ Useful for problems with mutually exclusive categories

Training Softmax Regression

The model is trained by minimizing the cross-entropy cost:

$$J(\Theta) = -\frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_k^{(i)} \log(\hat{p}_k^{(i)})$$

- ▶ Penalizes low probability for the true class
- ▶ When $K = 2$, this reduces to log loss
- ▶ No closed-form solution
- ▶ Train using Gradient Descent or another optimizer

Gradient with respect to class k :

$$\nabla_{\theta^{(k)}} J(\Theta) = \frac{1}{m} \sum_{i=1}^m (\hat{p}_k^{(i)} - y_k^{(i)}) \mathbf{x}^{(i)}$$

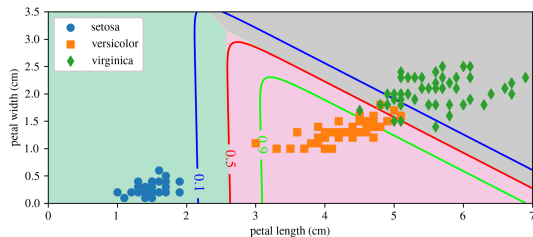
Example 5.4: Softmax Classification

The Iris dataset can be classified using petal length and petal width.

This example:

- ▶ Trains a multinomial logistic regression model
- ▶ Uses `multi_class="multinomial"`
- ▶ Requires a solver such as `lbfgs`

After training, a flower with 5 cm petals is classified as Iris-Virginica with probability 94.2%.



Softmax classification for the three Iris species.